EXPERIMENT – 2

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USER INTERFACE AND DESIGN

Implementation and Comparison of Command Line, Graphical, and Voice Interfaces for a Unit Conversion System

Features (same in all interfaces)

Convert:

- Kilometers ↔ Meters
- Kilograms ↔ Grams
- Celsius ↔ Fahrenheit

1. COMMAND LINE INTERFACE

```
def convert(value, unit):
if unit == "km-m":
return value * 1000
if unit == "m-km":
return value / 1000
if unit == "kg-g":
return value * 1000
if unit == "g-kg":
return value / 1000
if unit == "c-f":
return (value * 9/5) + 32
if unit == "f-c":
return (value - 32) * 5/9

print("Unit Converter")
print("Options: km-m, m-km, kg-g, g-kg, c-f, f-c")

value = float(input("Enter value: "))
unit = input("Enter conversion type: ").lower()
result = convert(value, unit)
```

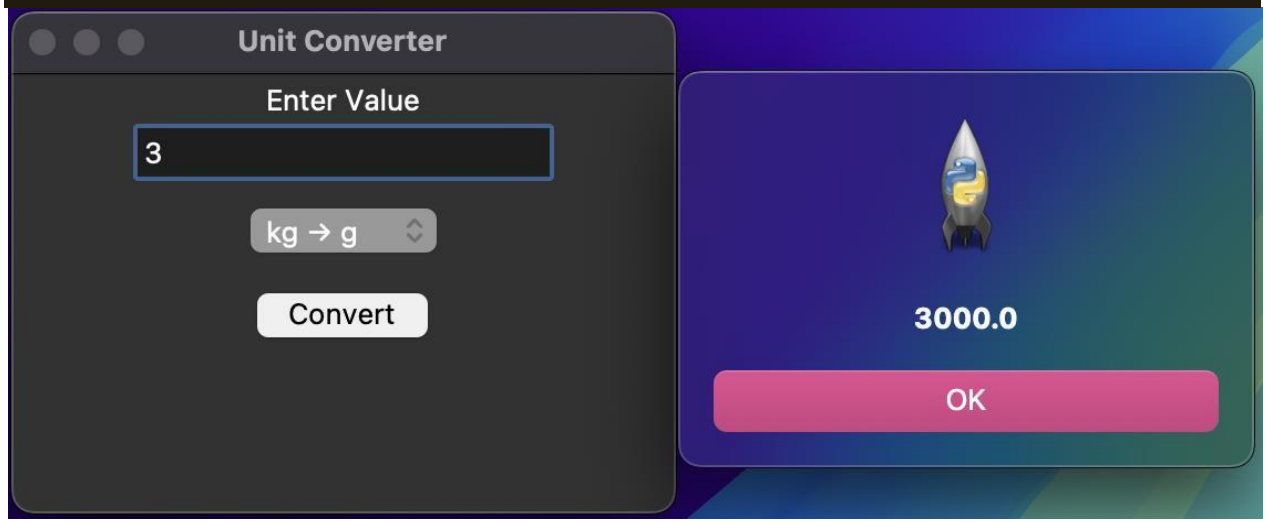
```
print("Result:", round(result, 2))
```

```
feminnadass@Feminnas-MacBook-Air Python % /usr/local/bin/python3.14 /Volumes/opstorage/Projects/Python/vendor_gui.py
Unit Converter
Options: km-m, m-km, kg-g, g-kg, c-f, f-c
Enter value: 3
Enter conversion type: kg-g
Result: 3000.0
```

2. GRAPHICAL USER INTERFACE

```
import tkinter as tk
from tkinter import messagebox


def convert():
try:
value = float(entry.get())
unit = option.get()


if unit == "km → m":
result = value * 1000
elif unit == "m → km":
result = value / 1000
elif unit == "kg → g":
result = value * 1000
elif unit == "g → kg":
result = value / 1000
elif unit == "°C → °F":
result = (value * 9/5) + 32
elif unit == "°F → °C":
result = (value - 32) * 5/9


messagebox.showinfo("Result", f"{round(result,2)}")


except:
messagebox.showerror("Error", "Enter valid number")


root = tk.Tk()
root.title("Unit Converter")
root.geometry("300x200")


tk.Label(root, text="Enter Value").pack()
entry = tk.Entry(root)
entry.pack()


option = tk.StringVar()
option.set("km → m")


tk.OptionMenu(root, option,
"km → m", "m → km",
"kg → g", "g → kg",
"°C → °F", "°F → °C").pack(pady=10)


tk.Button(root, text="Convert", command=convert).pack()
```

```
root.mainloop()
```



3. VOICE USER INTERFACE

```
import speech_recognition as sr

def process(command):
words = command.lower().split()

try:
value = float([w for w in words if w.replace('.', '', 1).isdigit()][0])

if "kilometer" in words or "km" in words:
print(value * 1000, "meters")

elif "meter" in words:
print(value / 1000, "kilometers")

elif "kilogram" in words or "kg" in words:
print(value * 1000, "grams")

elif "gram" in words:
print(value / 1000, "kilograms")

elif "celsius" in words:
print((value * 9/5) + 32, "fahrenheit")

elif "fahrenheit" in words:
print((value - 32) * 5/9, "celsius")
```

```
except:
print("Could not understand command")

r = sr.Recognizer()

with sr.Microphone() as source:
print("Speak your conversion command...")
audio = r.listen(source)

try:
text = r.recognize_google(audio)
print("You said:", text)
process(text)
except:
print("Speech not recognized")
```

```
feminnadass@Feminnas-MacBook-Air Python % /usr/local/bin/python3.14 /Volumes/opstorage/Projects/Python/vendor_vui.py
Speak your conversion command...

3
kg-g
```

4. USER SATISFACTION COMPARISON:

```
cli = int(input("CLI rating (1-5): "))
gui = int(input("GUI rating (1-5): "))
vui = int(input("VUI rating (1-5): "))

avg = (cli + gui + vui) / 3
print("Average Satisfaction:", round(avg,2))
```

```
feminnadass@Feminnas-MacBook-Air Python % /usr/local/bin/python3.14 /Volumes/opstorage/Projects/Python/survey.py
CLI rating (1-5): 5
GUI rating (1-5): 5
VUI rating (1-5): 1
Average Satisfaction: 3.67
```